

GRADUATION THESIS – PROJECT PROPOSAL

1. Student Information

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2. Project Title

AI Study Buddy – Personalized Learning Assistant Powered by Artificial Intelligence

3. Project Objectives

- To develop a mobile full-stack application that helps students study more effectively.
- To utilize Artificial Intelligence (AI) for summarizing documents, generating quizzes, suggesting study schedules, and providing an intelligent learning assistant. 
- To create a personalized learning environment that helps users track their progress and optimize study habits.

4. Scope and Main Features

Target Users: University students and self-learners.

Main Features:

1. Manage subjects, assignments, deadlines, and notes.
2. AI Study Chatbot: ask questions, **summarize content**, and generate quizzes from study materials.
3. **OCR & NLP integration: extract text from images or PDF files.** 
4. Learning Analytics Dashboard: visualize progress and suggest optimal study times.
5. Google Authentication & Cloud Storage using Firebase / PostgreSQL.

5. Technologies

Component	Technology
Mobile App	React Native
Backend API	FastAPI (Python) / Node.js (NestJS)
Database	PostgreSQL / Firebase Firestore
AI & NLP	OpenAI API / Gemini API
OCR	Tesseract OCR / Google Vision API
Deployment	Render / Vercel / Firebase Hosting

6. Expected Outcomes

- A fully functional mobile application built with React Native and a modern, user-friendly interface.
- Integrated AI assistant capable of answering questions and summarizing learning materials.
- Data analytics features to monitor learning progress and generate visual reports.
- A complete thesis report describing the system architecture, APIs, and implementation process.

7. Future Development

- Add voice-based interaction with the AI assistant.
- Build a community platform for sharing study materials.
- Integrate machine learning models to automatically suggest optimal study times.
- Publish the app on Google Play / App Store.

8. Deliverables

- Mobile app APK demo.
- Source code (GitHub repository).
- Full thesis report (PDF format).

Student: Nguyen Tan Thinh

Expected Duration: October 2025 – June 2026

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